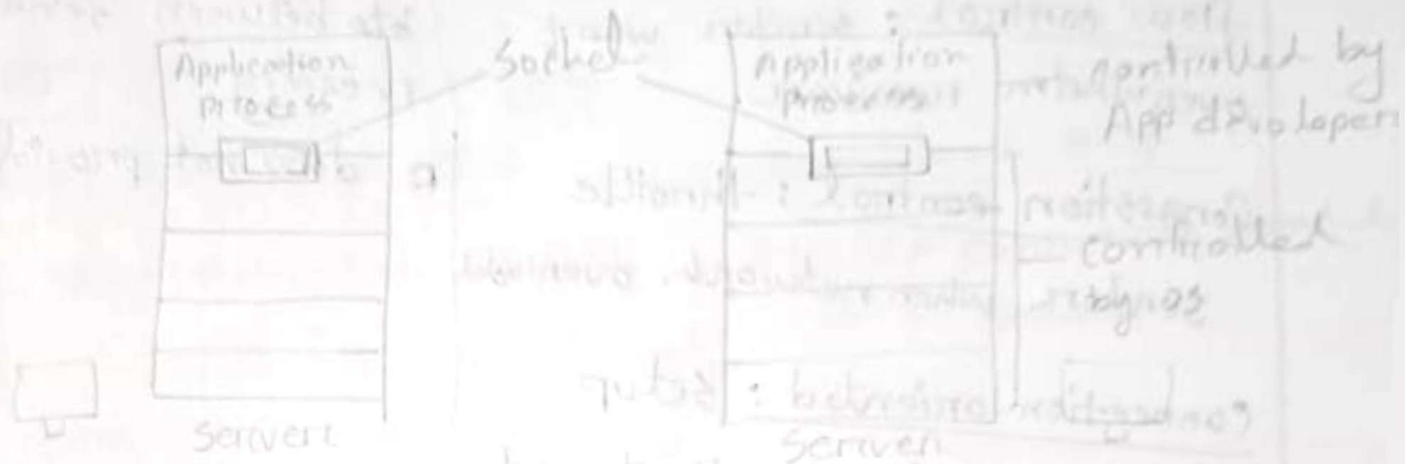


Sockets:

- process sends/receives messages to/from its socket



Addressing processes

- ↳ to receive messages, process must have identifier
- ↳ host device has unique 32 bit IP address.

Q. does IP address of host on which process runs suffice for identifying the process?

⇒ No, many processes can be running on same host

Socket → IP address and port number combination

Internet transport protocols service:

TCP service

reliable transport between sending and receiving process
flow control: sender won't overwhelm receiver.

Congestion control: throttle sender when network overloaded

Connection-oriented: setup required between client and server processes.

does not provide: timing, minimum throughput guarantee, ~~some~~ security.

UDP Service

■ Unreliable data transfer between sending and receiving

■ does not provide

Web and HTTP

Web page consists of object → HTML file,

JPEG image, Java

www.somerschool.edu / someDept / pie.gif

host name

Path name

HTTP overview

HTTP uses TCP.

client initiates. --

HTTP connection: two types

non-persistent HTTP

1. TCP connection opened
2. at most one object sent over TCP connection

persistent HTTP

1. TCP connection opened to a server

Non persistent HTTP:

example:

Non persistent HTTP: response time

~~RTT~~ Non persistent HTTP response time = $2RTT + \text{file transmission time}$

HTTP request message

□ two types of HTTP messages: request, response

□ HTTP request format: General format

□:

Other HTTP request message

→ Post method

→ Head method

→ GET method

→ Put method